

CANCER PRESENTATION

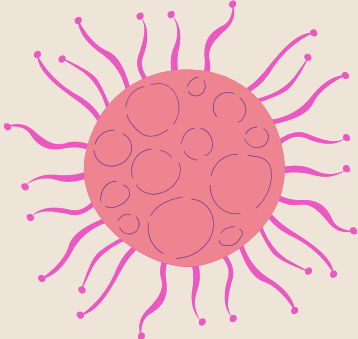
Pan-Cancer Classification from Gene Expression Data



PROJECT OVERVIEW

- **Objective:** Build a machine learning system that classifies cancer type based on a patient's gene expression profile
- **Problem type:** Multi-class classification (33 TCGA tumor types)
- **Why it matters:** Gene expression patterns are highly cancer-specific – a reliable classifier can support diagnostic support tools, biomarker discovery, and precision oncology research
- **Approach:** Combine dimensionality reduction (PCA), class grouping, hierarchical "expert" models, and ensemble/SVM classifiers to achieve high accuracy across highly imbalanced, high-dimensional data

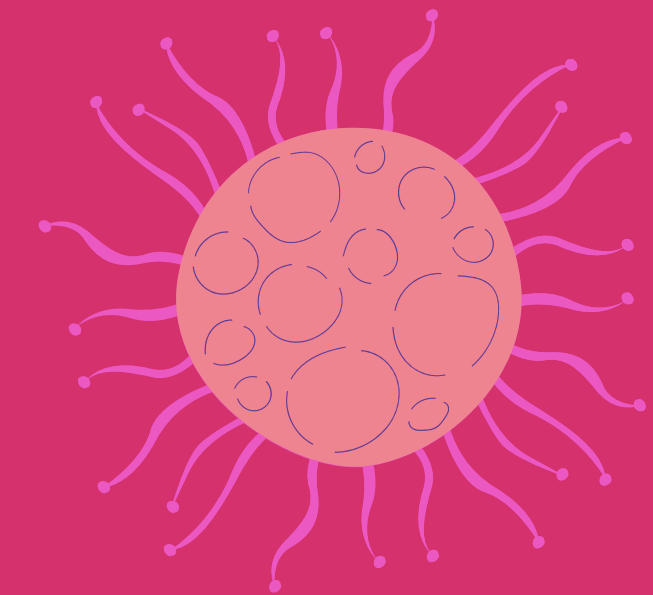
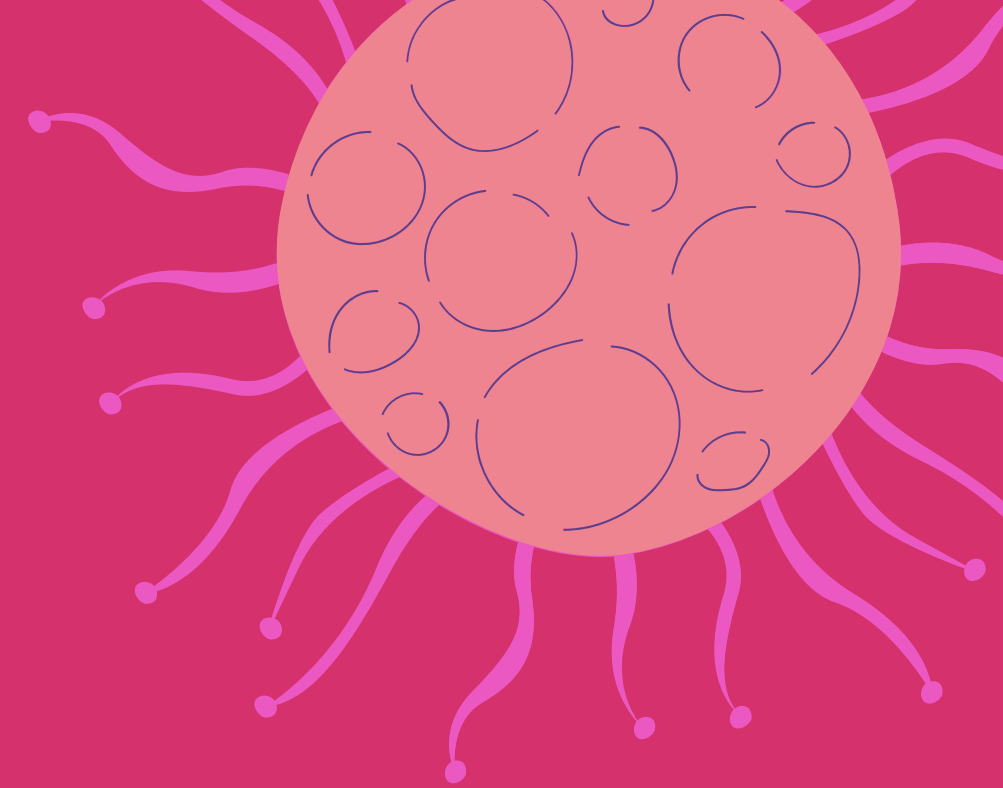
OVERVIEW



Item	Detail
Source	UCSC Xena, TCGA Pan-Cancer (PANCAN)
Patients	11,069 (gene expression file)
Genes	20,531
Cancer types	33
Files used	• Gene Expression Matrix: EB++AdjustPANCAN_IlluminaHiSeq_RNASeqV2.geneExp.xena • Phenotype/Survival Labels: Survival_SupplementalTable_S1_20171025_xena_sp.txt
Final merged dataset	11,014 patients
Key step	Datasets merged via inner join on sampleID



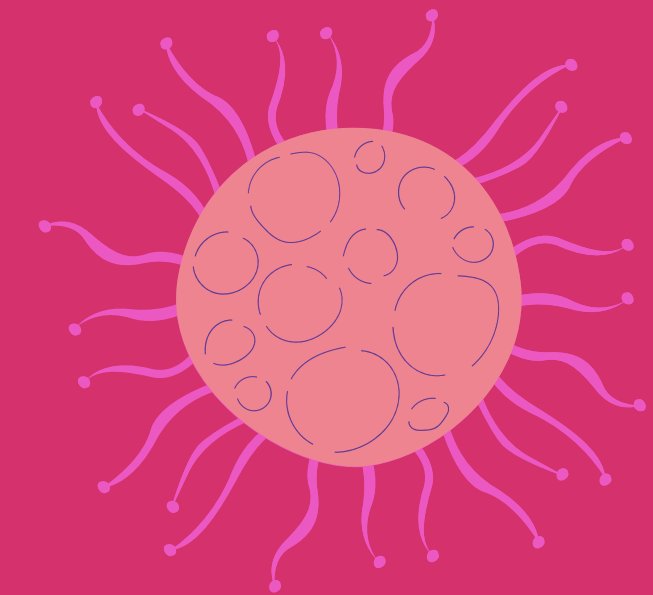
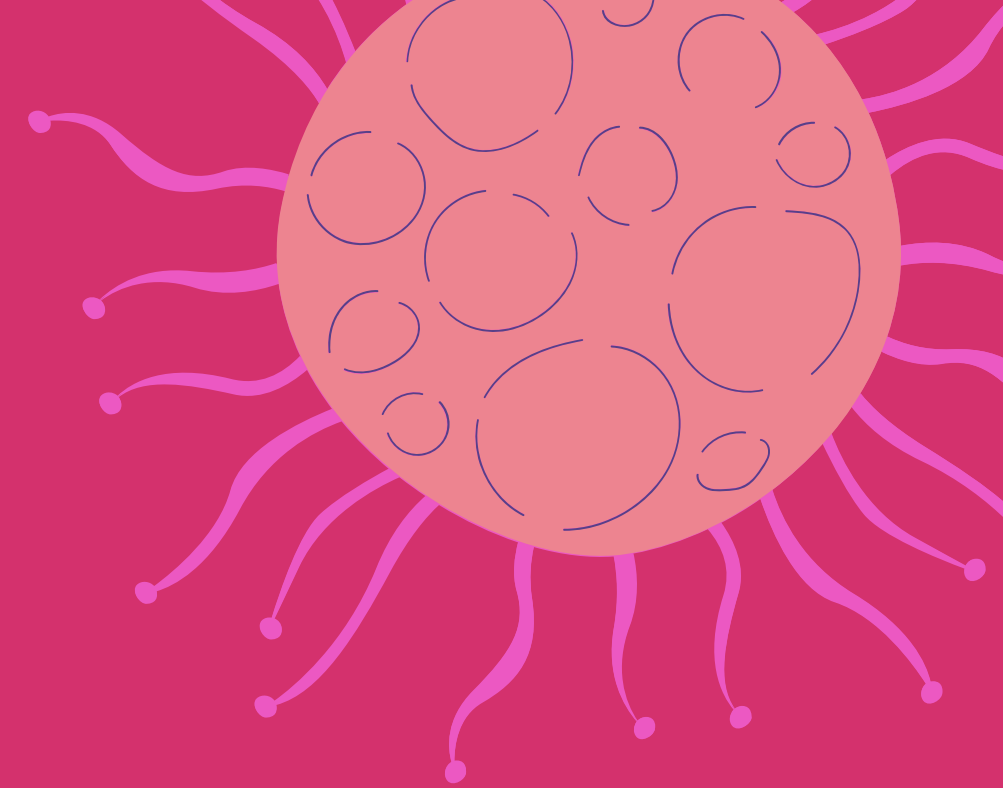
Data Preprocessing



PREPROCESSING & DIMENSIONALITY REDUCTION

- **Missing values:** Filled with 0 (unexpressed genes)
- **Duplicates:** Checked and confirmed none present
- **Standardization:** StandardScaler applied $\rightarrow$ mean $\approx$ 0, std $\approx$ 1
- **Variance Thresholding:** Removed near-zero-variance genes to cut noise
- **PCA:** Reduced from 20,530 genes $\rightarrow$ 500 principal components
- **Label Encoding:** Cancer type strings converted to numeric labels
- **Class Grouping:** Merged biologically similar subtypes into parent categories:
 - COAD + READ $\rightarrow$ COLORECTAL
 - LUAD + LUSC $\rightarrow$ LUNG_CANCER

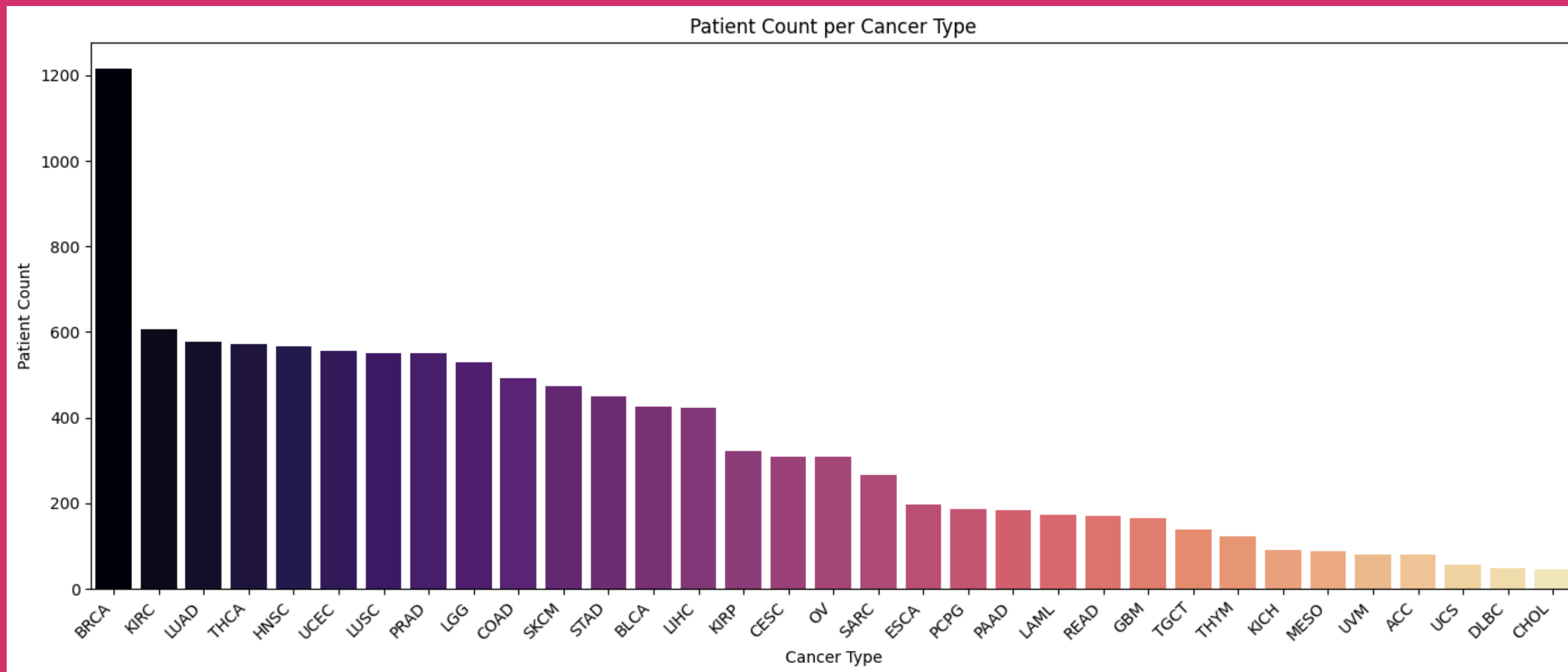
Exploratory Data Analys

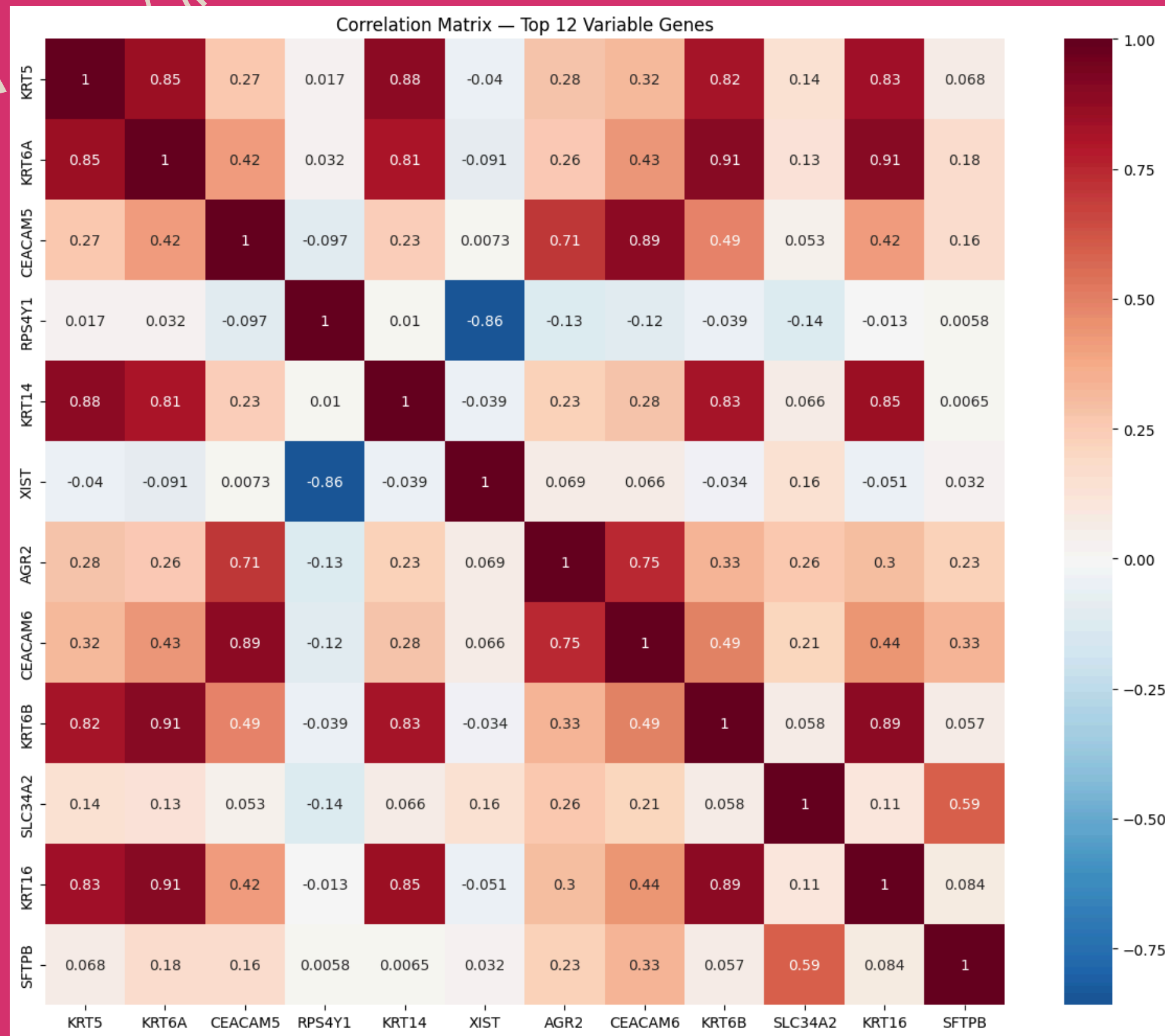




- The dataset is imbalanced — some cancer types (e.g., BRCA) have far more samples than rare types
- This imbalance directly informed later modeling decisions: `class_weight='balanced'` and SMOTE oversampling

CLASS DISTRIBUTION
PATIENTS PER CANCER
TYPE





GENE CORRELATION STRUCTURE



- Several of the most variable genes are highly correlated with one another
- This indicates redundancy in the feature space, supporting the need for dimensionality reduction before modeling



- Each cancer type has a distinct gene expression signature
- the top genes in BRCA are completely different from the top genes in LUAD.

Top 10 Expressed Genes: BRCA vs LUAD Comparison Table				
	Top Genes in BRCA	BRCA Mean Exp	Top Genes in LUAD	LUAD Mean Exp
0	ACTB	16.432230	ACTB	16.813484
1	EEF1A1	16.295404	FTL	16.310625
2	COL1A1	16.227825	SFTPB	16.152639
3	ACTG1	16.099216	EEF1A1	16.113346
4	COL1A2	15.885059	ACTG1	16.064060
5	FN1	15.838951	CD74	16.061405
6	COL3A1	15.767916	ADAM6	15.946880
7	EEF2	15.453402	B2M	15.847239
8	GAPDH	15.443437	TPT1	15.817623
9	TPT1	15.410956	FN1	15.604835

GENE EXPRESSION
COMPARISON INSIGHT



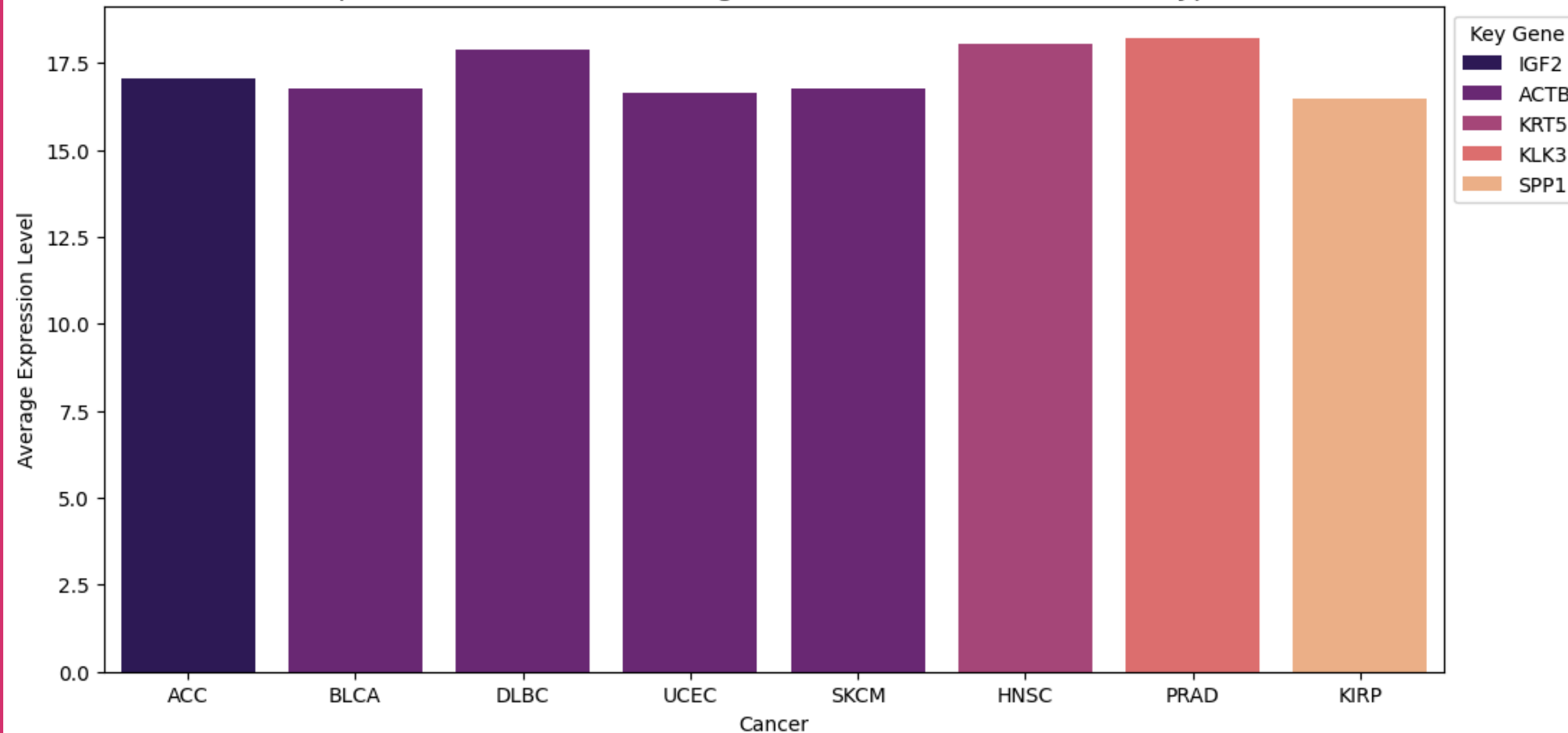
- This confirms that gene expression is a biologically meaningful basis for classification, not just a statistical artifact.

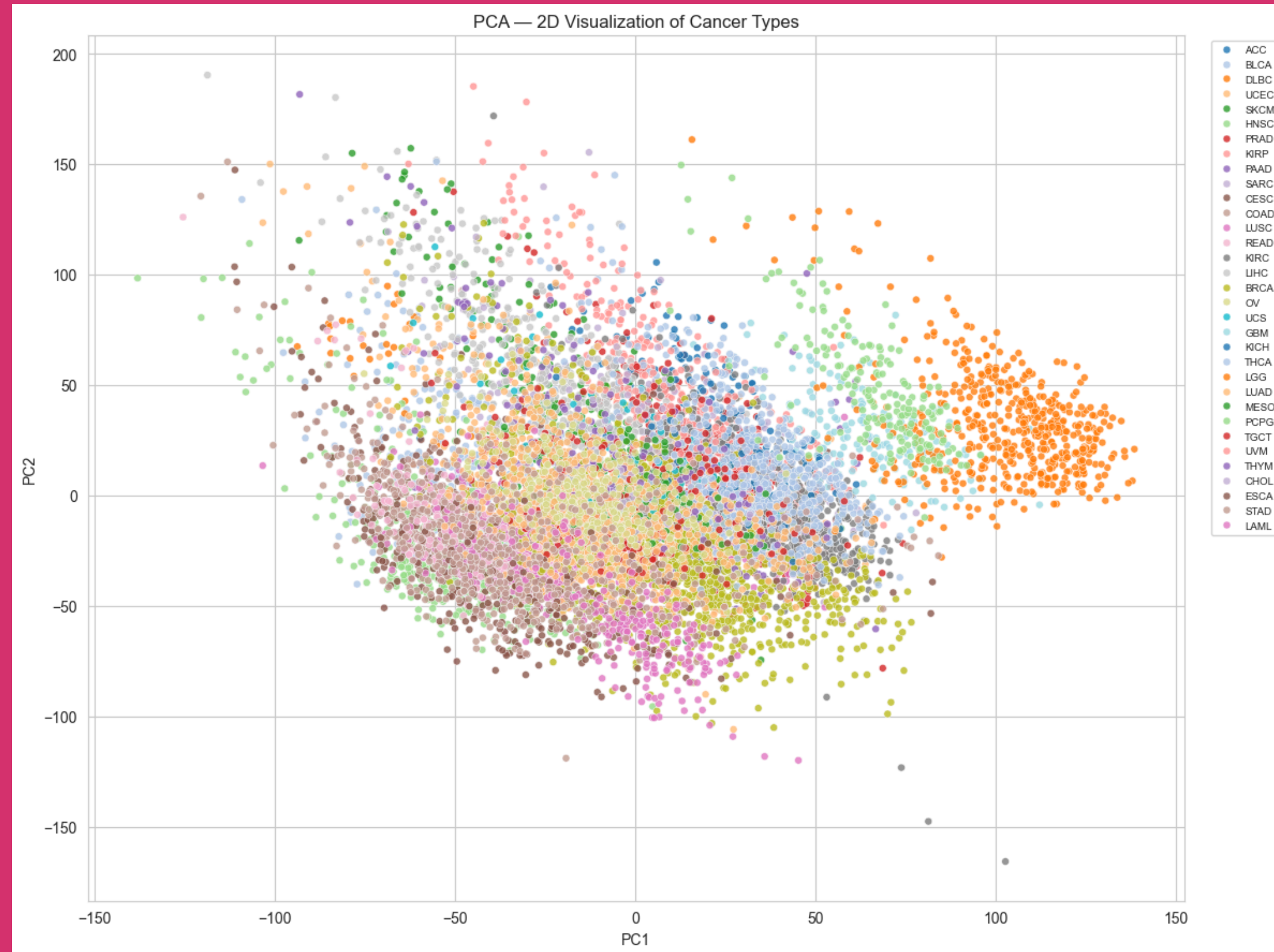


- Each cancer type has a distinct dominant gene (e.g., **BRCA** → **TFF1/PRAME**, **PRAD** → **KLK3**)
- Confirms that gene expression is biologically meaningful for classification, not random noise

CANCER-SPECIFIC BIOMARKERS

Specific Biomarkers: The "Signature Gene" for each Cancer Type





GENE CORRELATION STRUCTURE

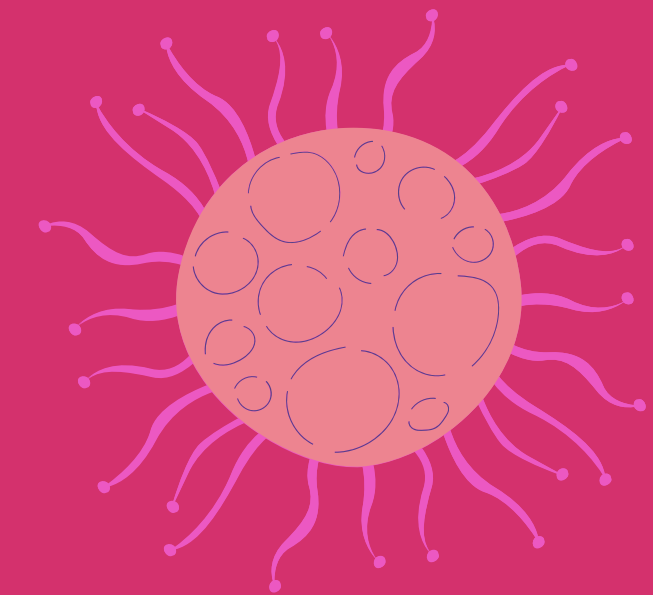
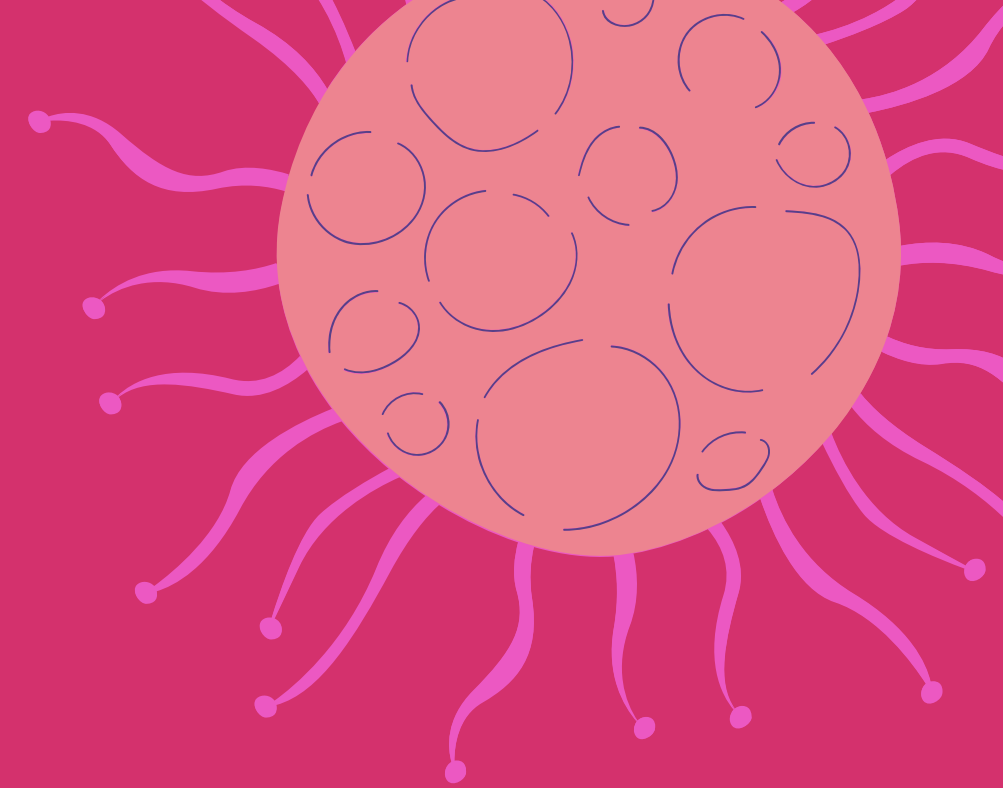


- Even in just 2 principal components, samples cluster clearly by cancer type
- Strong visual evidence that gene expression profiles are separable — validating the classification approach

CHALLENGES & SOLUTIONS

Challenge	Detail
20,000+ genes — too high-dimensional to visualize or model directly	Applied variance threshold filtering + PCA to reduce to meaningful dimensions
Imbalanced cancer type samples (some types far more represented)	Identified imbalance early via distribution analysis; accounted for it in model evaluation
Static notebook plots aren't reusable by non-technical stakeholders	Rebuilt key insights as an interactive Power BI dashboard for live exploration

Model Selection



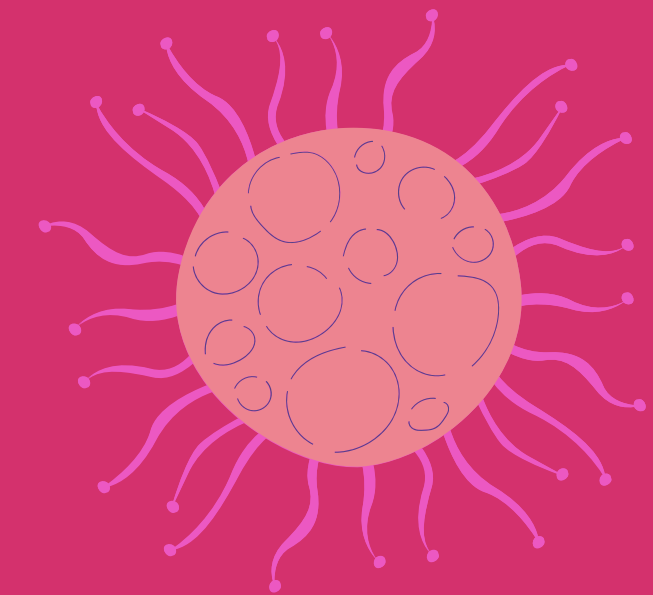
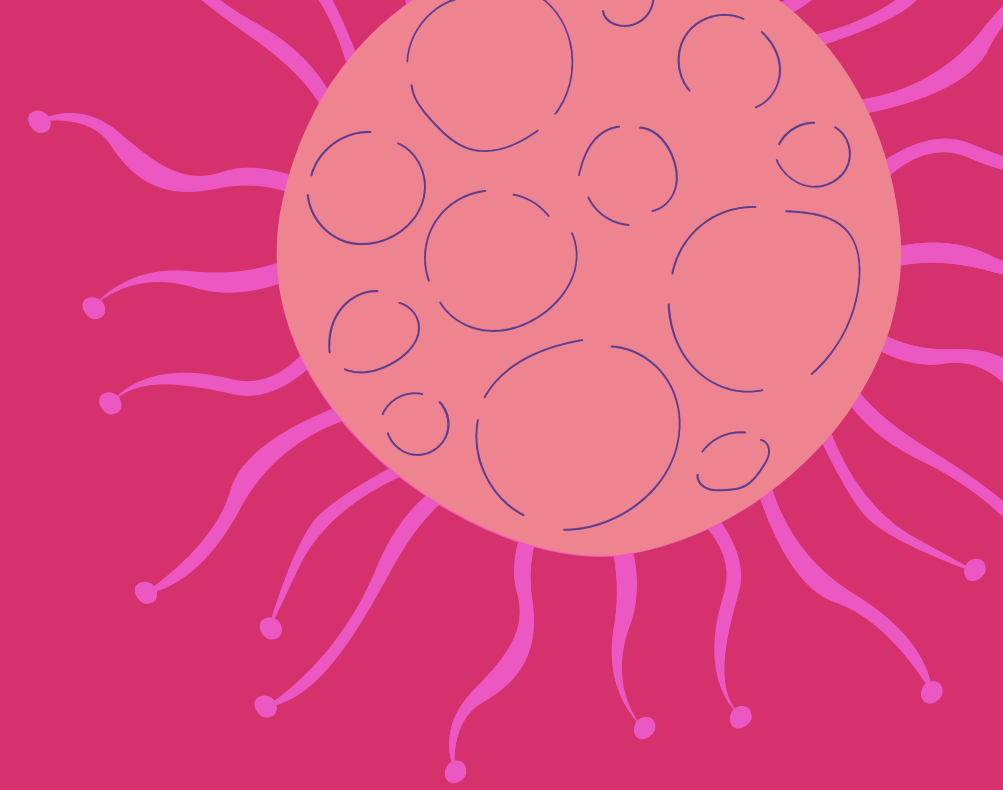
MODEL SELECTION

Two candidates, chosen for how well they fit this data — many features, relatively few samples, uneven class sizes.

Model	Why we tried it
Random Forest	Handles thousands of gene features naturally; ranks genes by importance — useful for spotting biomarkers
SVM (linear)	Known to perform well when there are far more features than samples, as with gene expression data

Both models were trained with balanced class weights, since some cancer types have far more samples than others.

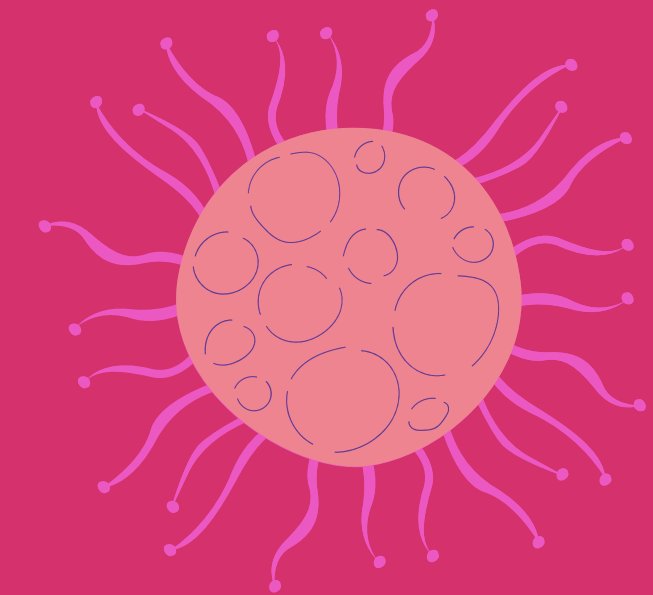
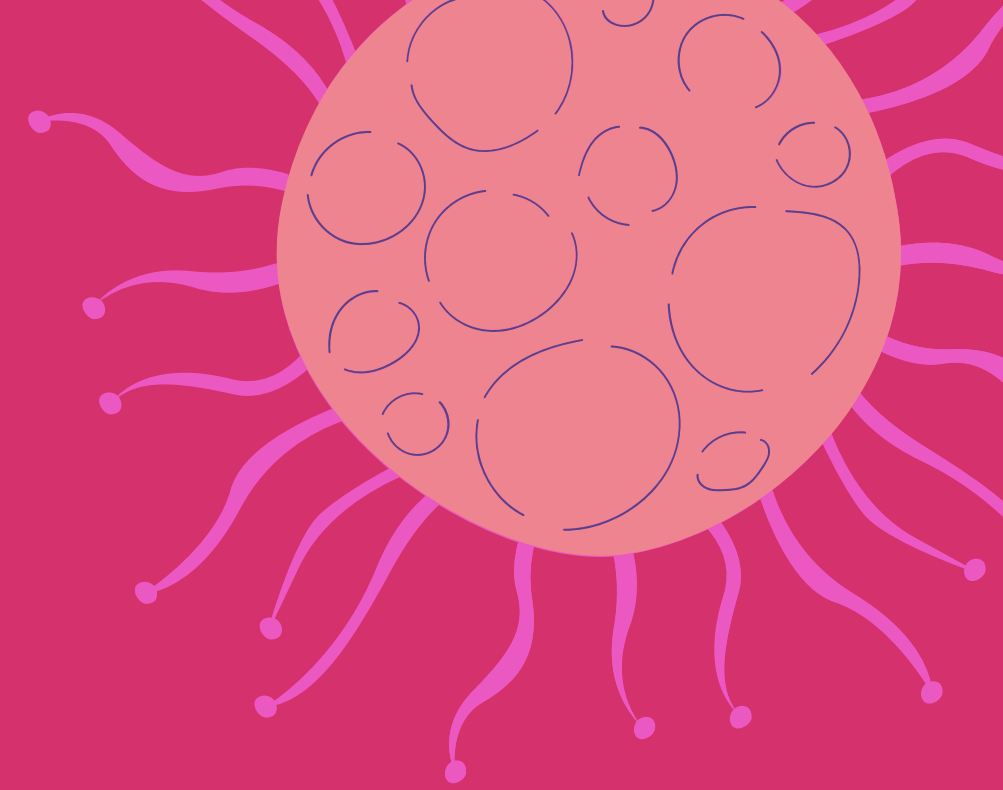
Training Process



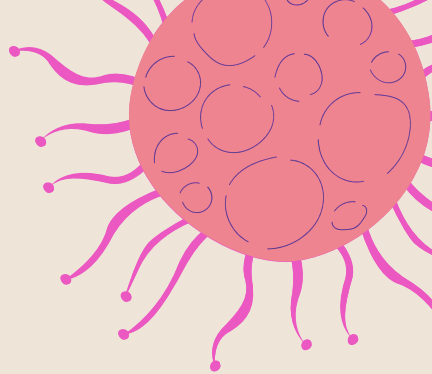
TRAINING PROCESS

Step	What happened
1. Reduce dimensions	20,000+ genes compressed with PCA before training
2. Split the data	80% train / 20% test, stratified by cancer type
3. Group similar types	COAD+READ → Colorectal; LUAD+LUSC → Lung Cancer
4. Train parent model	One classifier trained across all grouped categories
5. Add specialist experts	Extra focused models trained only on the hardest look-alike pairs

Evaluation



EVALUATION — OVERALL RESULTS



Model	Accuracy	Precision	Recall	F1-Score
Random Forest	94.3%	94.4%	94.3%	94.1%
SVM (linear)	97.3%	97.4%	97.3%	97.3%

5-Fold Cross-Validation (checks the numbers above aren't a lucky split)

Model	Mean Accuracy	Std. Dev.
Random Forest	94.1%	± 0.84%
SVM (linear)	96.8%	± 0.53%

SVM stays consistently higher and tighter across folds — the more reliable of the two.

EVALUATION — WHERE IT STRUGGLES

OVERALL ACCURACY LOOKS STRONG, BUT A FEW RARE CANCER TYPES ARE CONSISTENTLY WEAK FOR BOTH MODELS.

Cancer Type	Samples	RF F1-Score	SVM F1-Score
CHOL (bile duct)	9	50	62
UCS (uterine carcinosarcoma)	11	67	83
ESCA (esophageal)	39	51	86
KICH (kidney chromophobe)	18	87	82

Why: these are the smallest classes in the dataset (as few as 9 patients) — too little data for either model to learn a reliable pattern. SVM handles the shortage better than Random Forest, but neither is strong here.

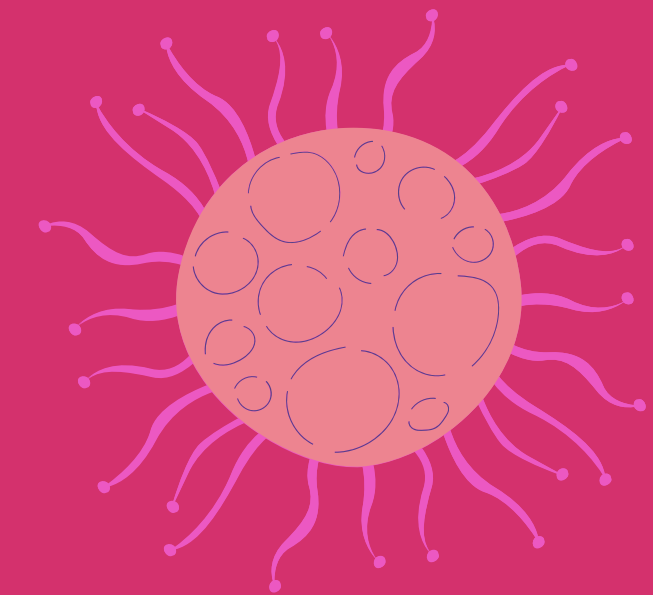
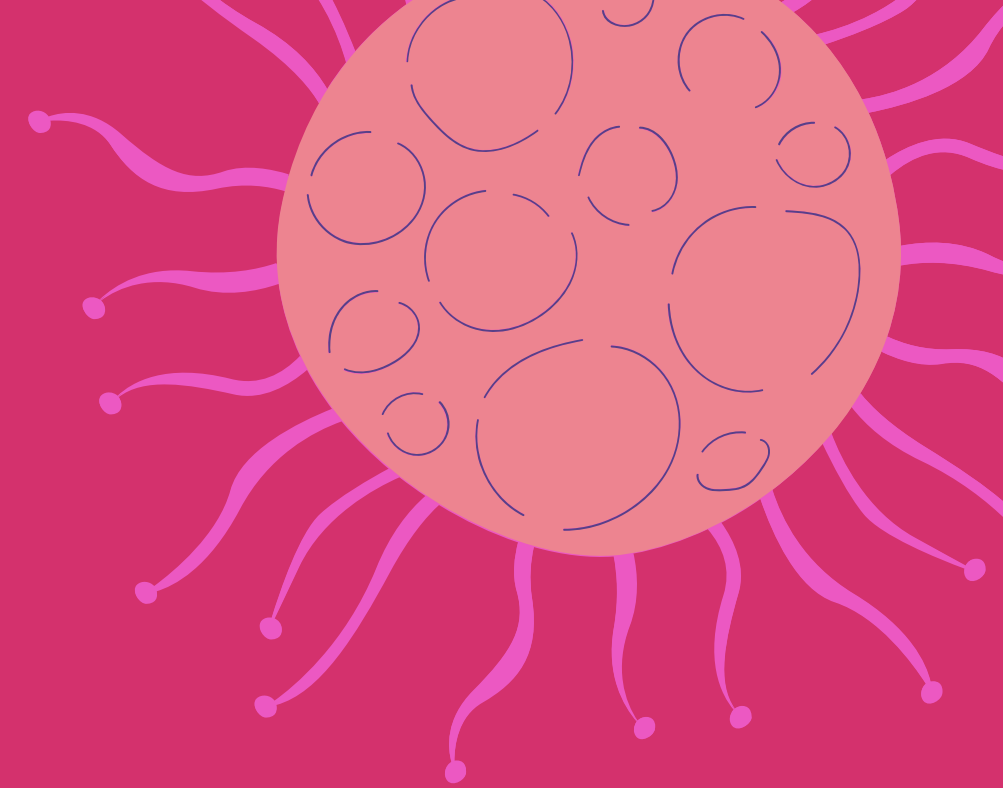
EVALUATION — SPECIALIST EXPERTS

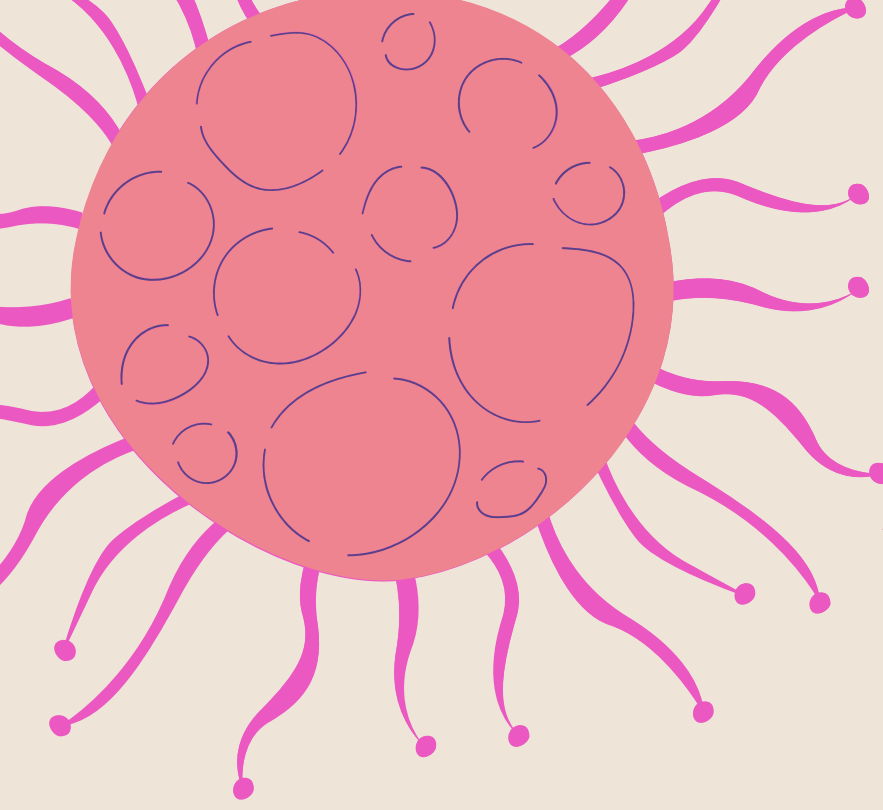
Extra models focused only on the two hardest look-alike pairs

Expert	Result
Colorectal (COAD vs READ) — first attempt	74% accuracy, but READ recall was 0% — the model never once caught a READ case
Colorectal — after feature selection + SMOTE	77% accuracy, READ F1 improved to 0.56 — better, still the weakest result in this project
Lung (LUAD vs LUSC)	92% accuracy, balanced across both subtypes — a clear success

COAD vs. READ remains the hardest distinction in the whole project — the two are molecularly very similar

Deployment





MODEL DEPLOYMENT

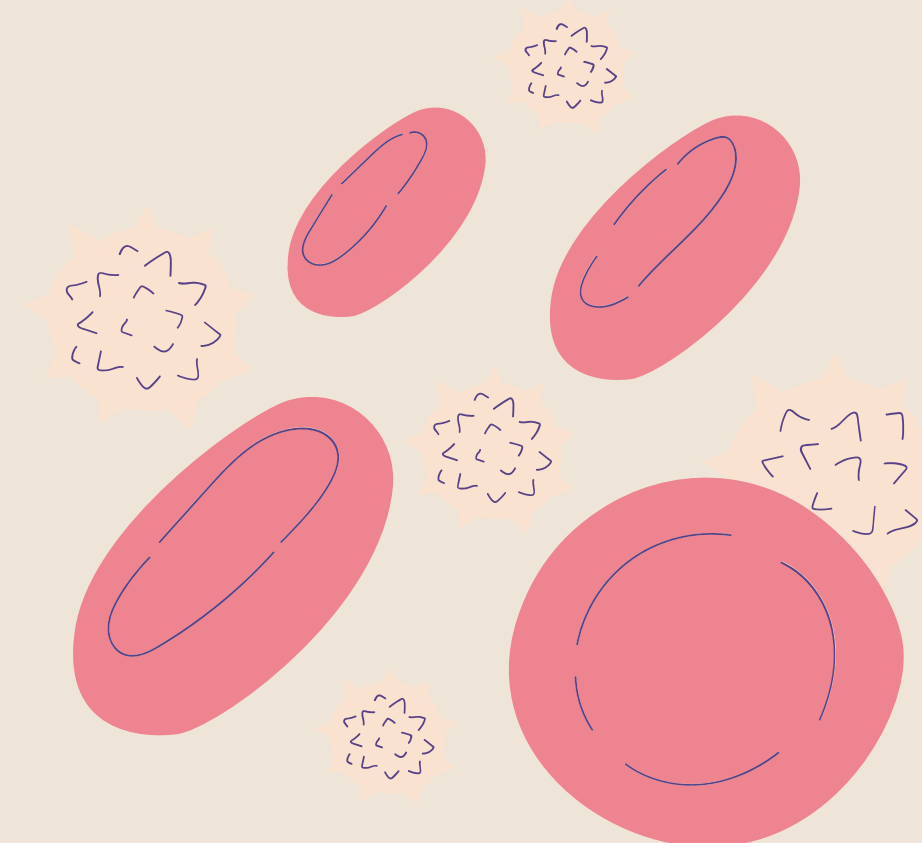
After selecting the best-performing model, the prediction workflow was prepared for deployment.

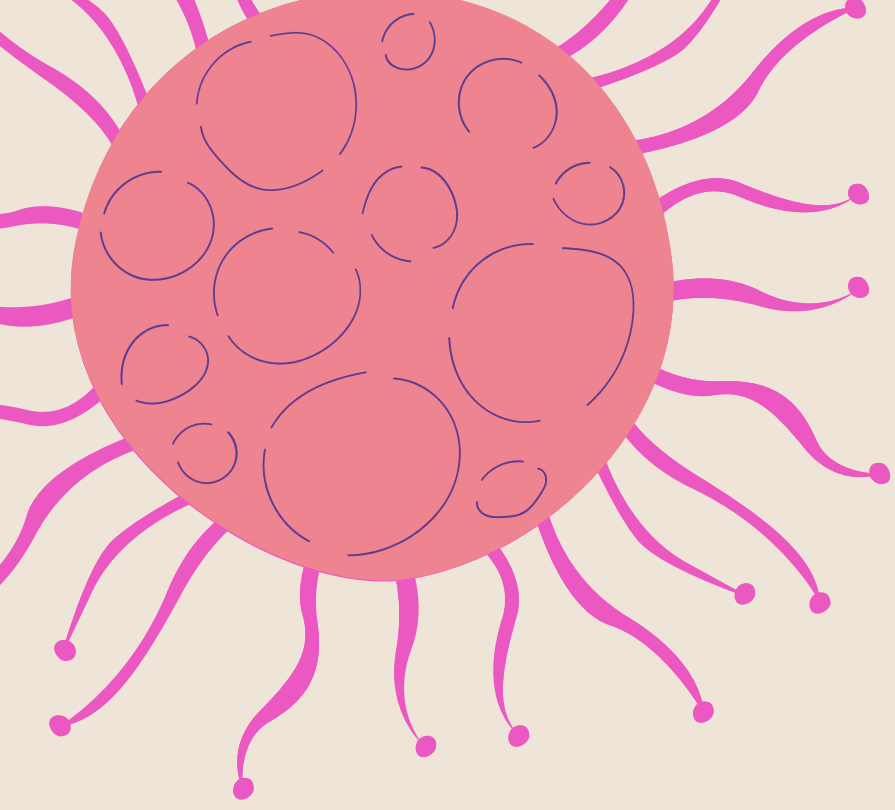
DEPLOYMENT PIPELINE

**Gene Expression Data → Data Preprocessing → PCA Transformation
→ Trained Linear SVM Model → Cancer Type Prediction → Prediction Dashboard**

DEPLOYMENT FEATURES

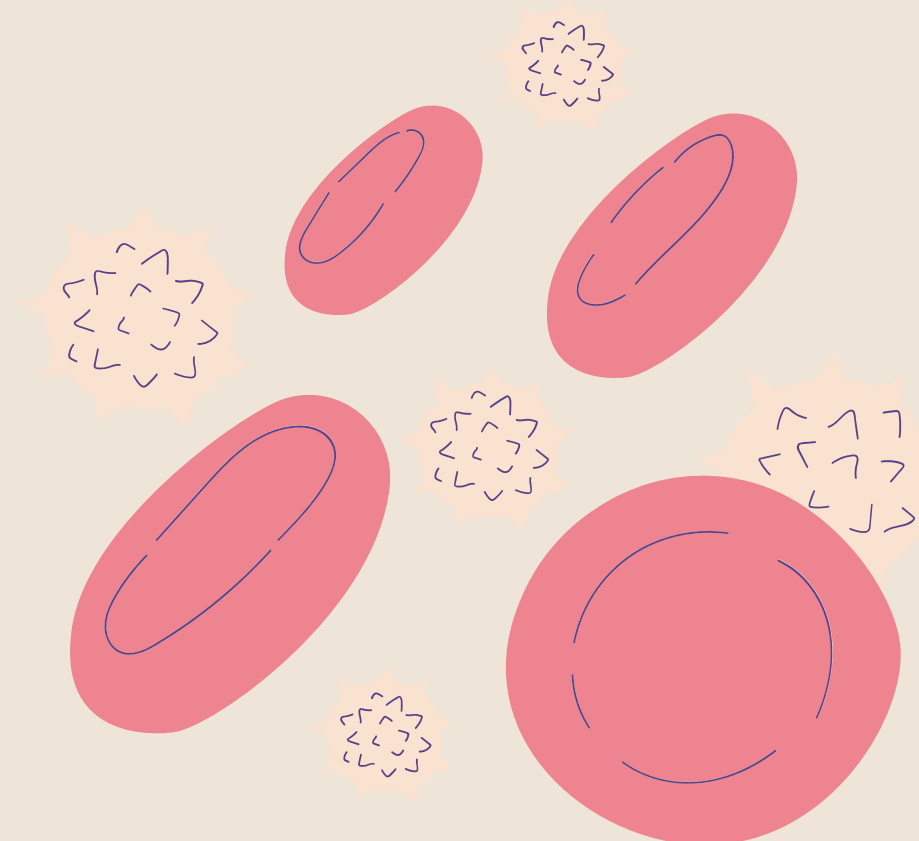
- Fast prediction for new patient samples.
- Consistent preprocessing before inference.
- Easy integration with clinical software.
- Interactive visualization using Power

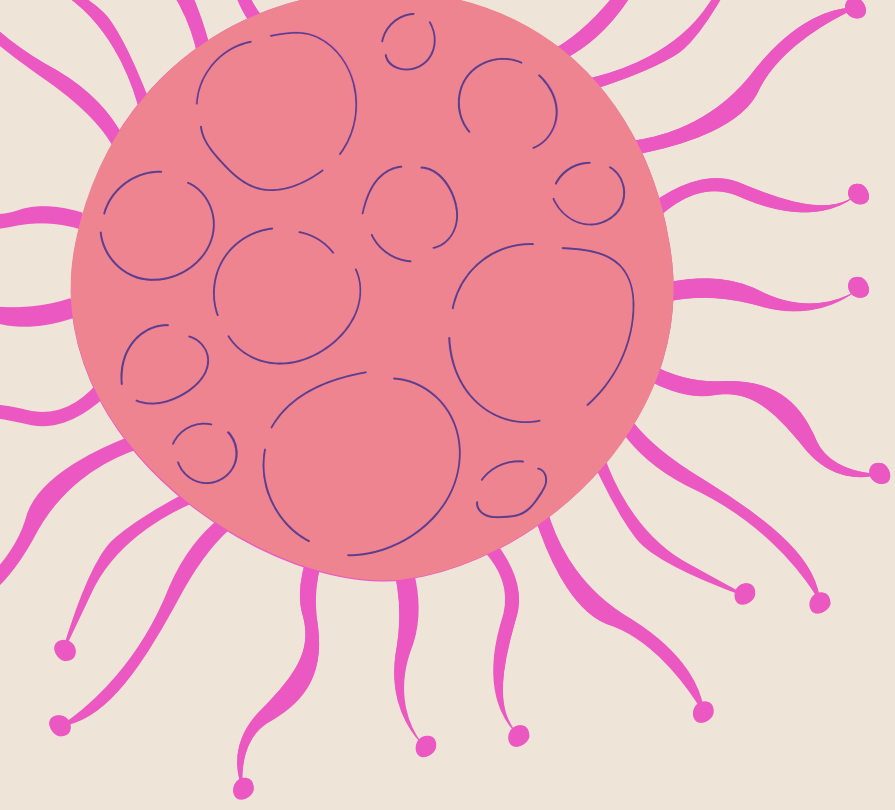




TRACKING EXPERIMENTS WITH MLFLOW

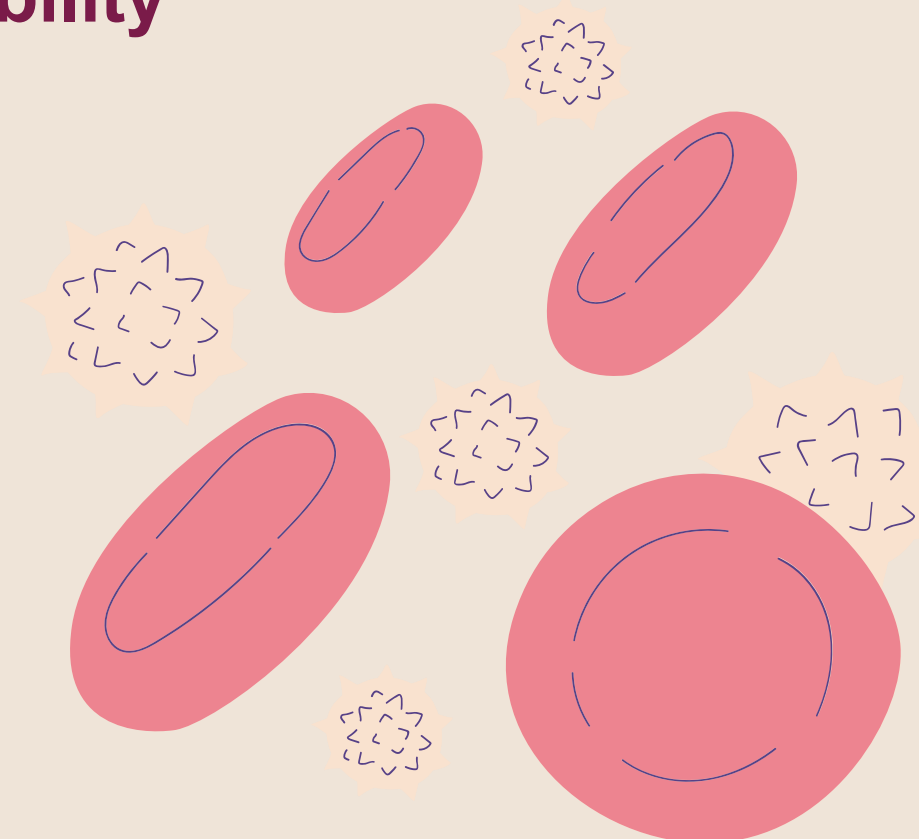
- Every model run (Random Forest, tuned SVM) logged with parameters, accuracy, F1-score, and the trained model artifact
- Enables comparing runs over time and reproducing any past result

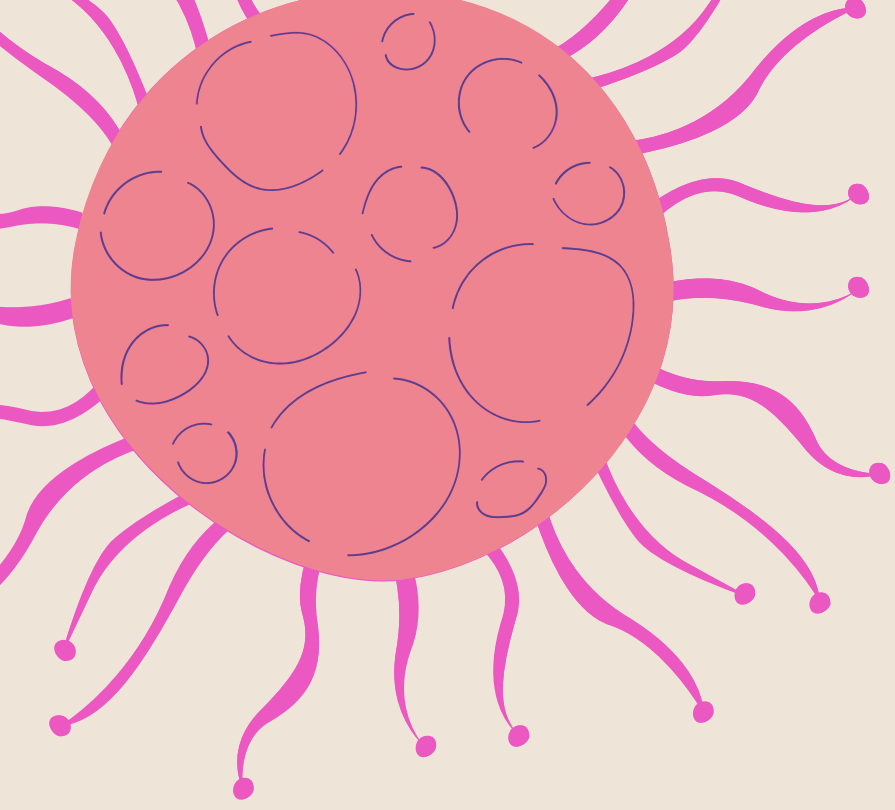




VERSIONED INFERENCE PIPELINE

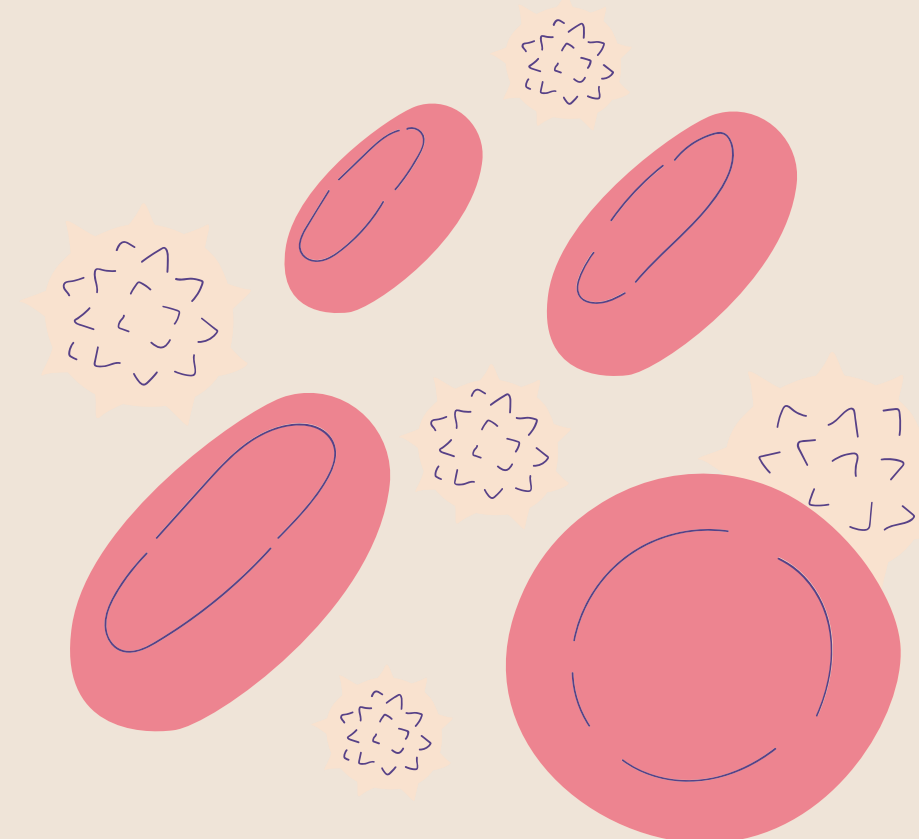
- Full pipeline (scaler + variance selector + PCA + label encoder + model) saved together as a versioned package (models/v<timestamp>/)
- Includes a metadata file (version, model type, accuracy, class list) for auditability

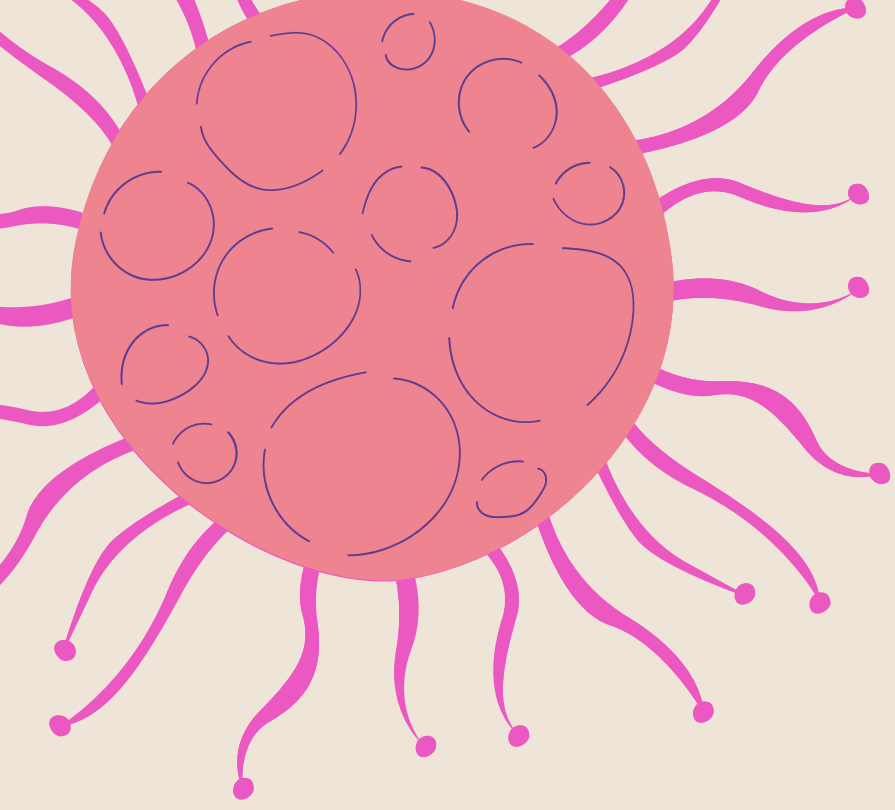




SERVING PREDICTIONS VIA REST API

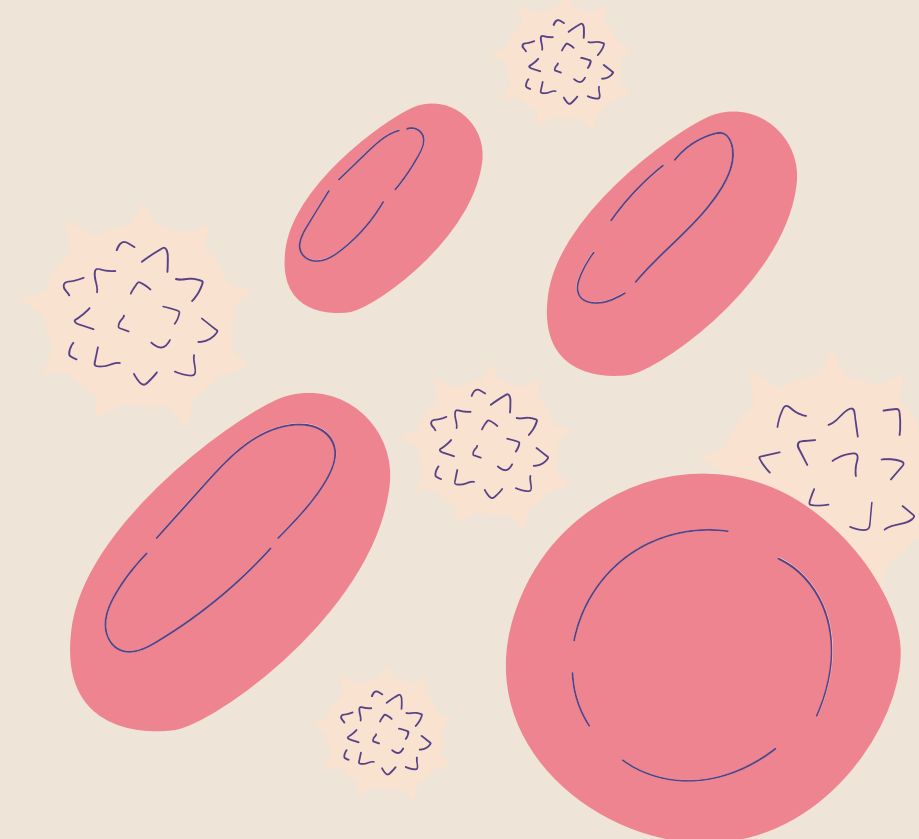
- Built a FastAPI service with two endpoints:
- /health — service status check
- /predict — accepts gene expression values, returns predicted cancer type + confidence score
- Deployable via uvicorn behind a real server/container

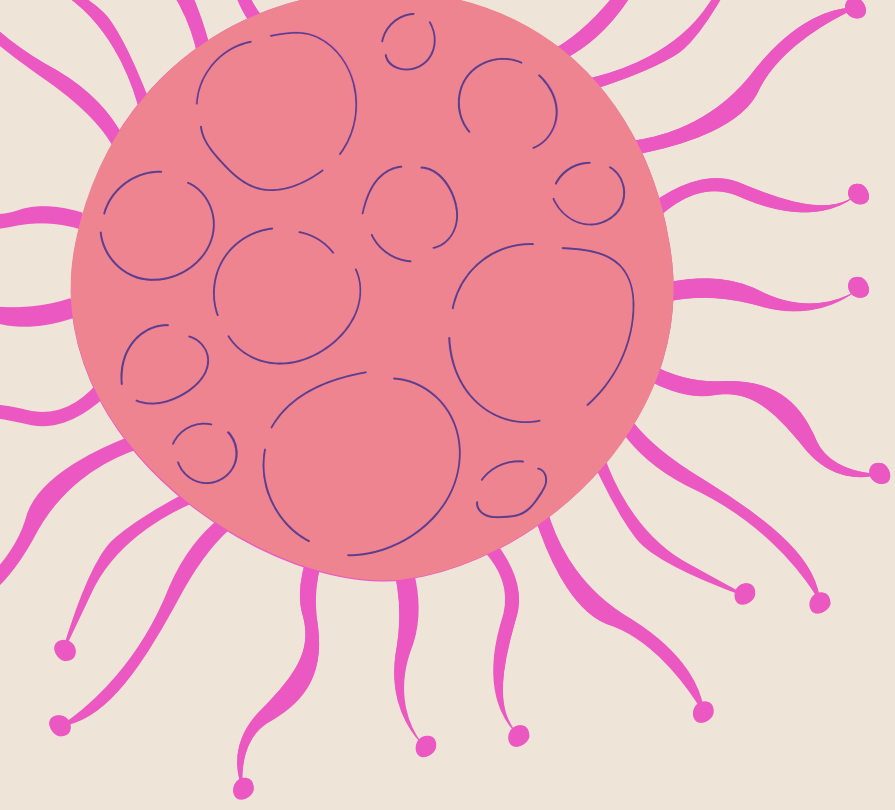




DETECTING DATA DRIFT IN PRODUCTION

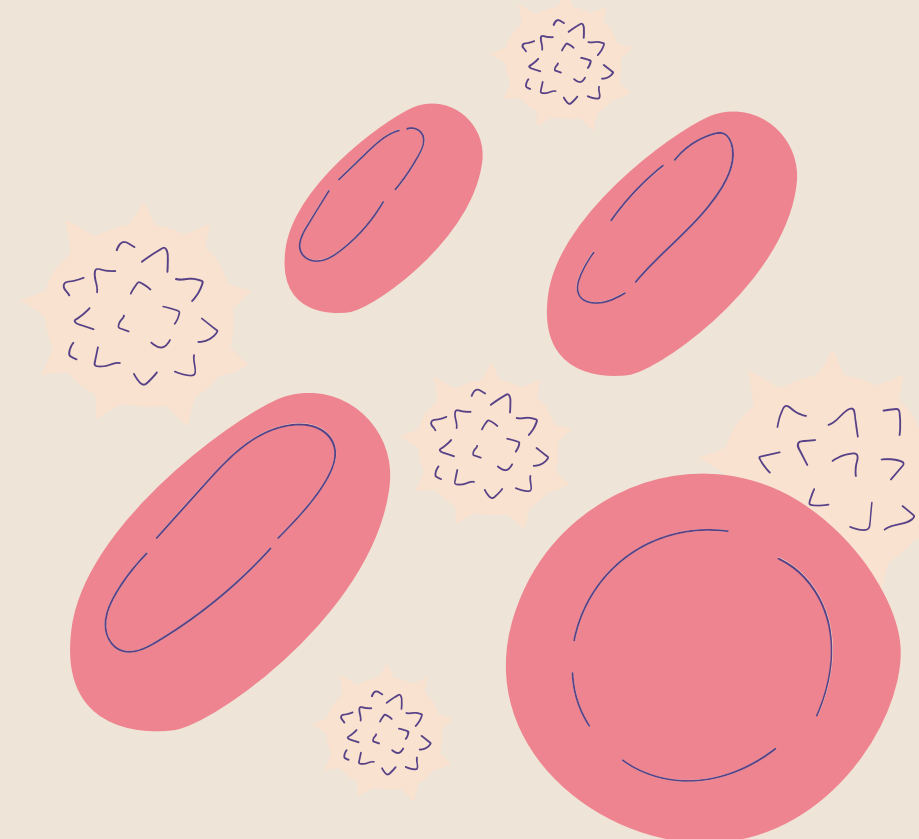
- Implemented Population Stability Index (PSI) to compare incoming data distribution against the training reference set, feature by feature
- Automatic alert triggers when PSI exceeds the industry-standard threshold (0.2), signaling the model may need retraining



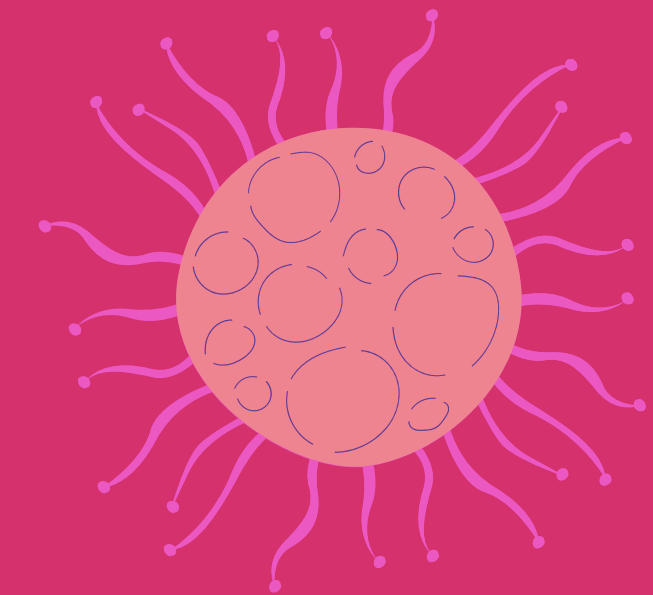
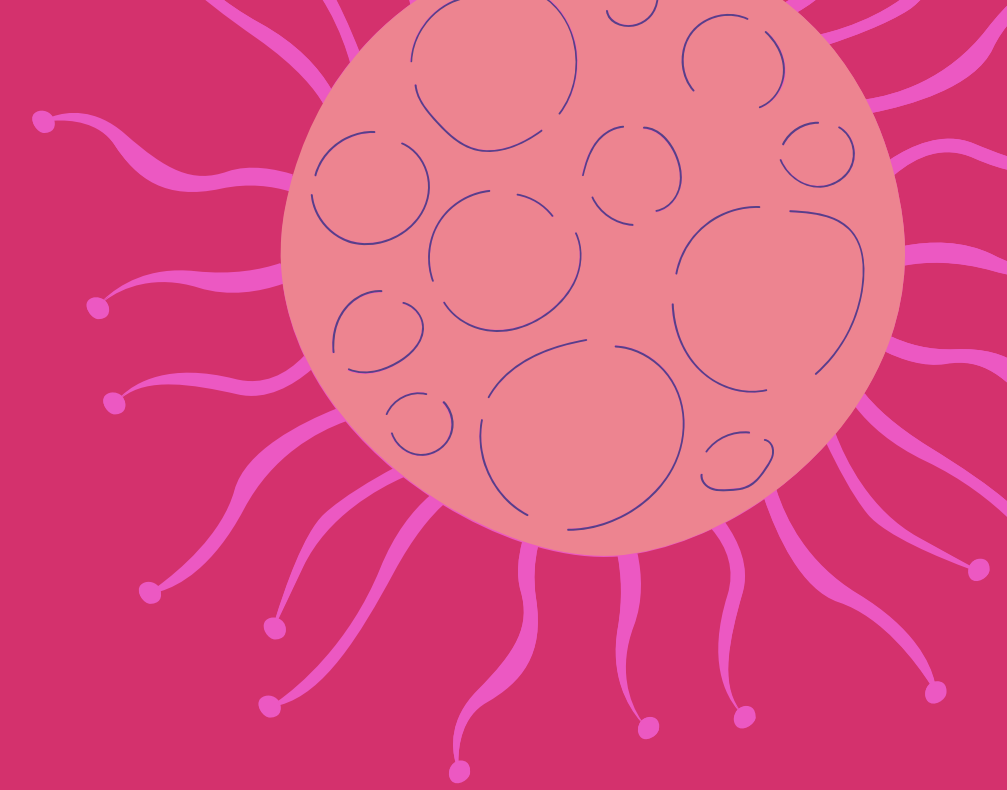


CONTINUOUS PERFORMANCE LOGGING

- A scheduled reporting function appends timestamped accuracy/F1 snapshots to a log file
- Designed to run on a recurring schedule (daily/weekly) against fresh incoming data to catch performance degradation early



Results and Conclusion

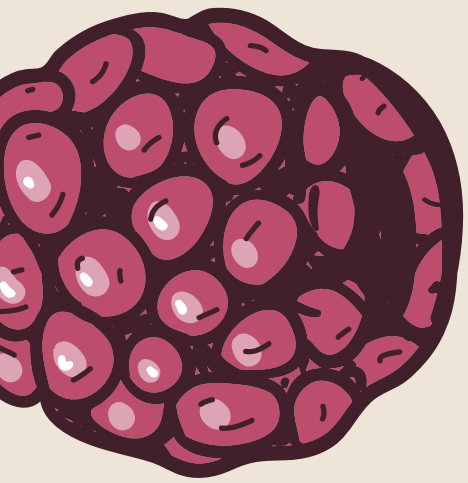


RESULTS & CONCLUSION



• RESULTS

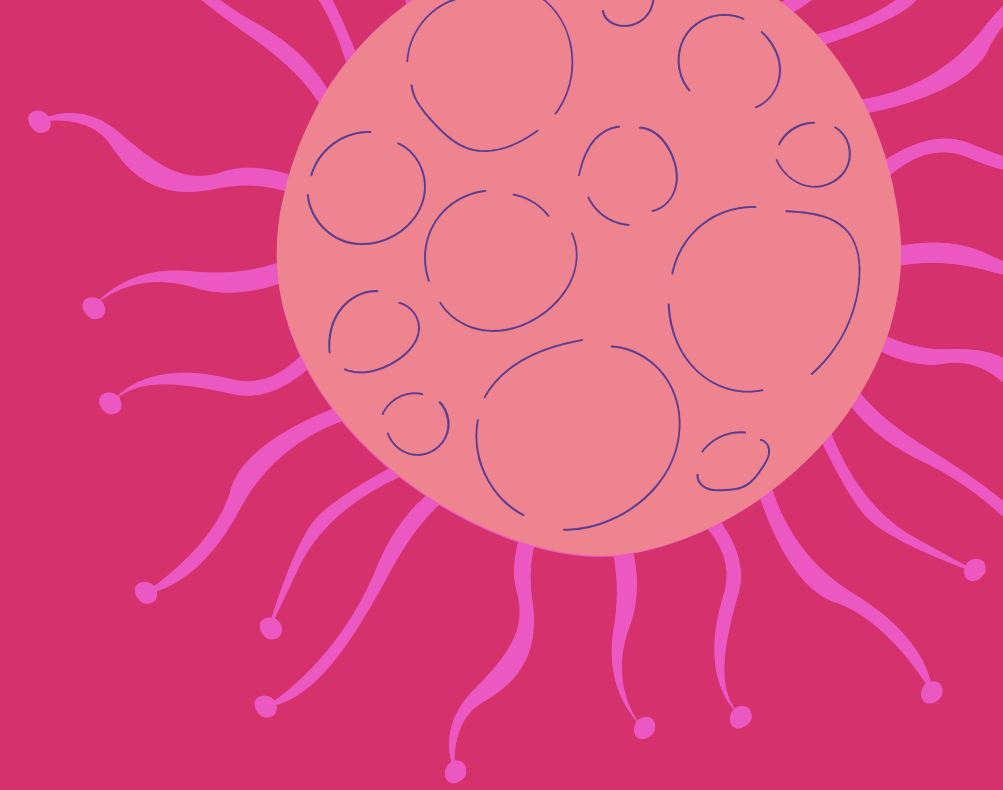
- Successfully classified 33 distinct types of cancer utilizing gene expression data.
- Reduced over 20,000 gene features into informative components through Principal Component Analysis (PCA).
- Developed an accurate and scalable machine learning pipeline.
- Achieved consistent performance across various cancer categories.
- Created an interactive dashboard to facilitate data exploration and result interpretation.



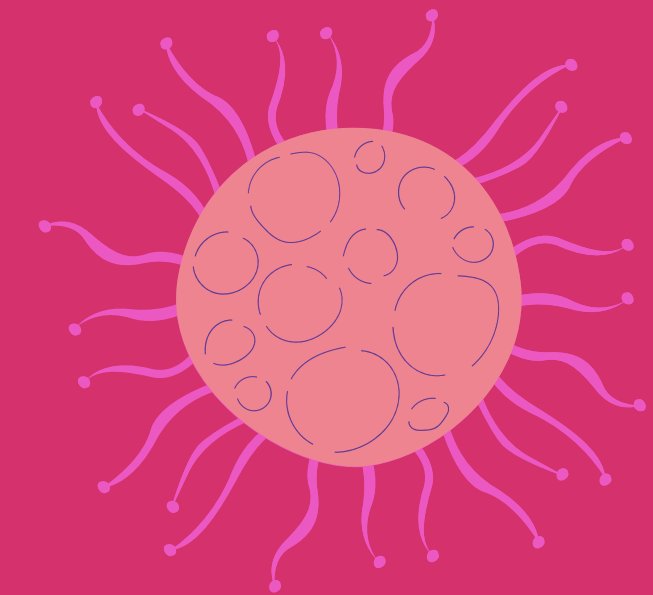
• CONCLUSION

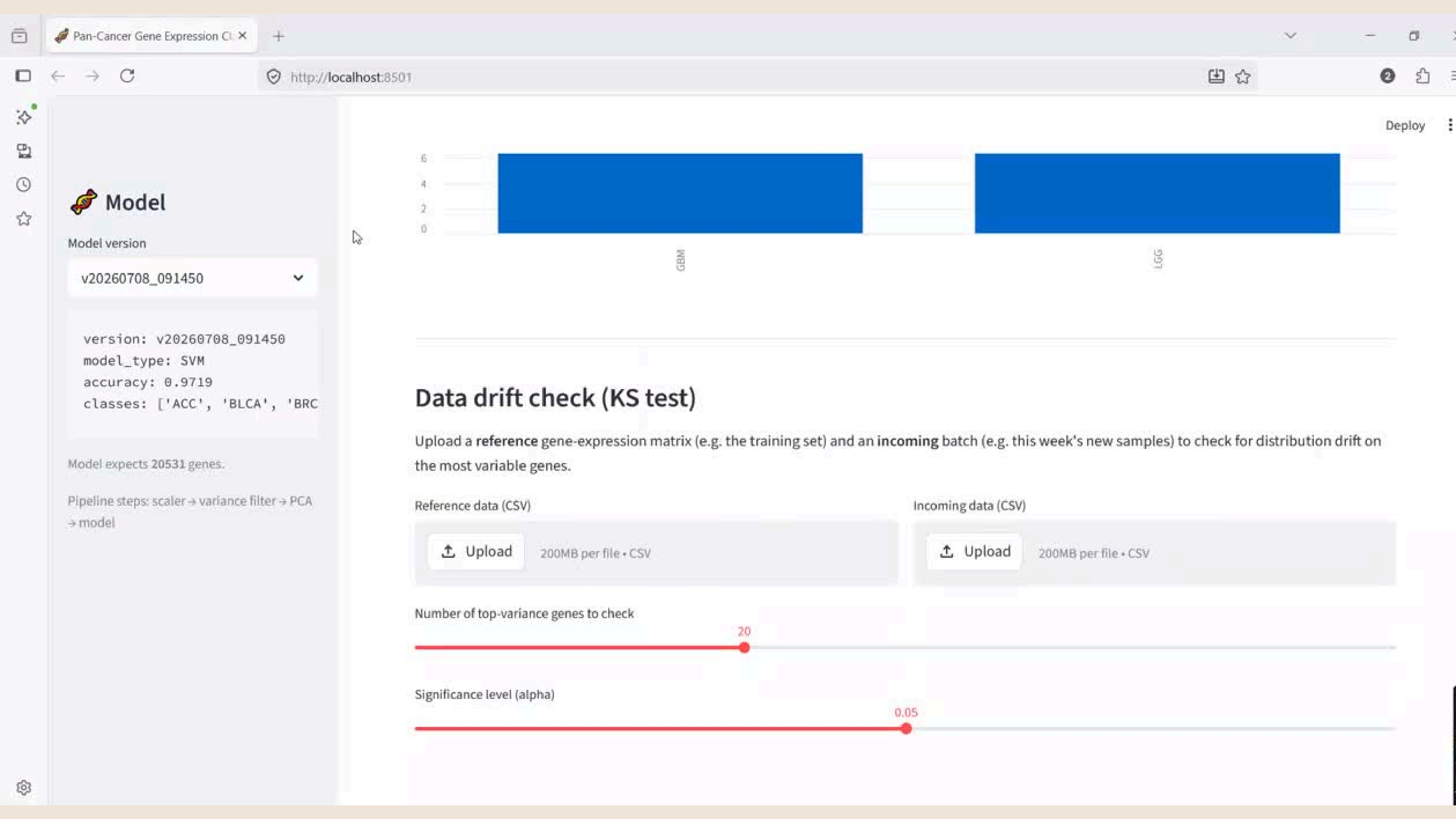
This project demonstrates that machine learning can effectively classify cancer types from gene expression profiles.

By combining data preprocessing, dimensionality reduction, and robust classification models, we developed an accurate prediction system that can support future precision medicine and clinical decision-making.



Live Demo





THANK YOU

